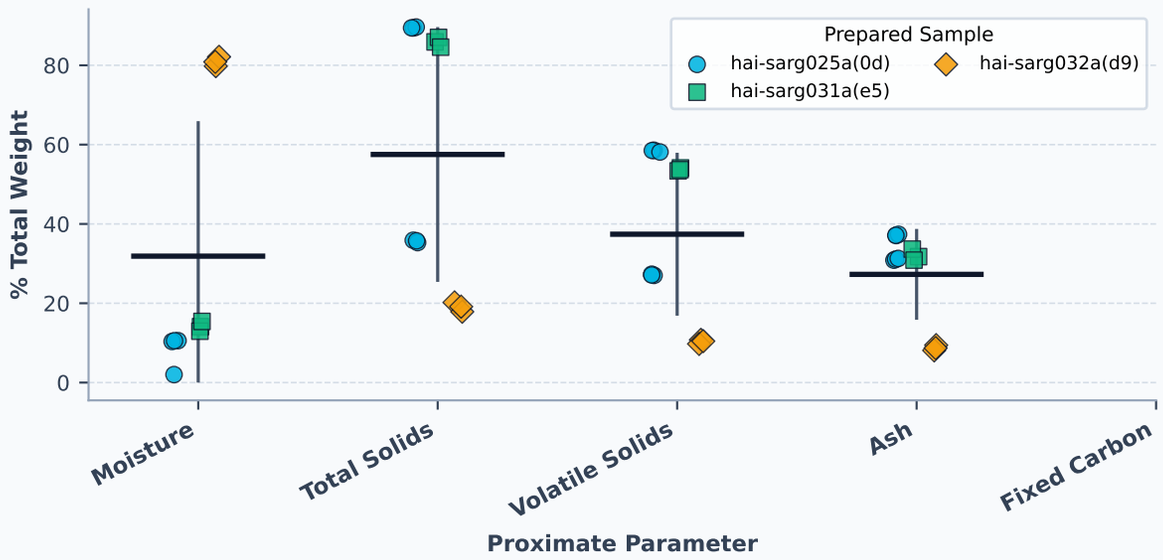
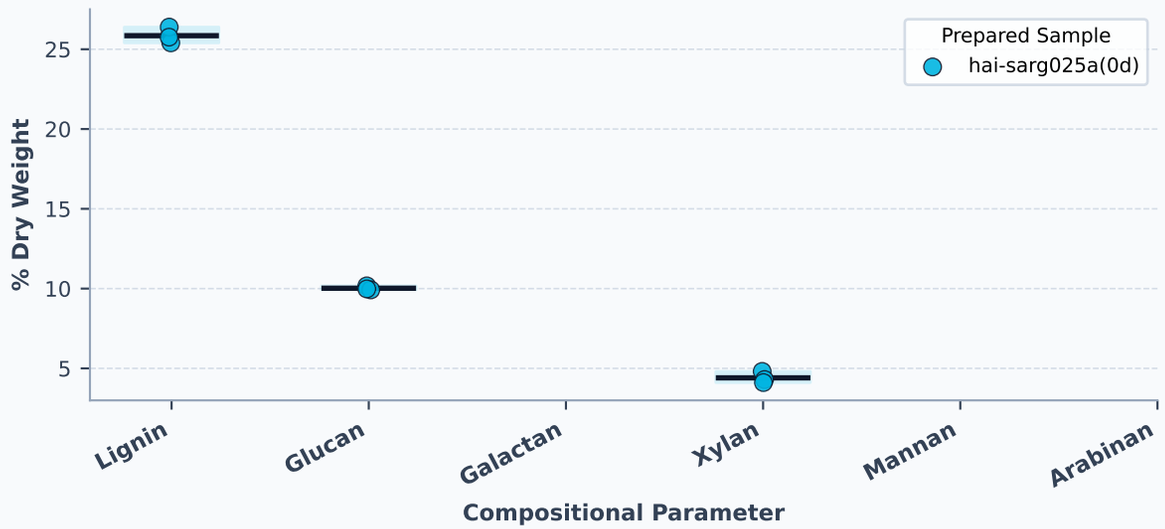
CA-BIOSITING SARGASSUM (MACROALGAE) — ANALYTICAL & BIOCONVERSION VISUAL DASHBOARD

252 Observations · 3 Field Samples (hai-sarg025, hai-sarg031, hai-sarg032) · 6 Prepared Samples · 243 QC Pass (96.4%) / 9 QC Fail

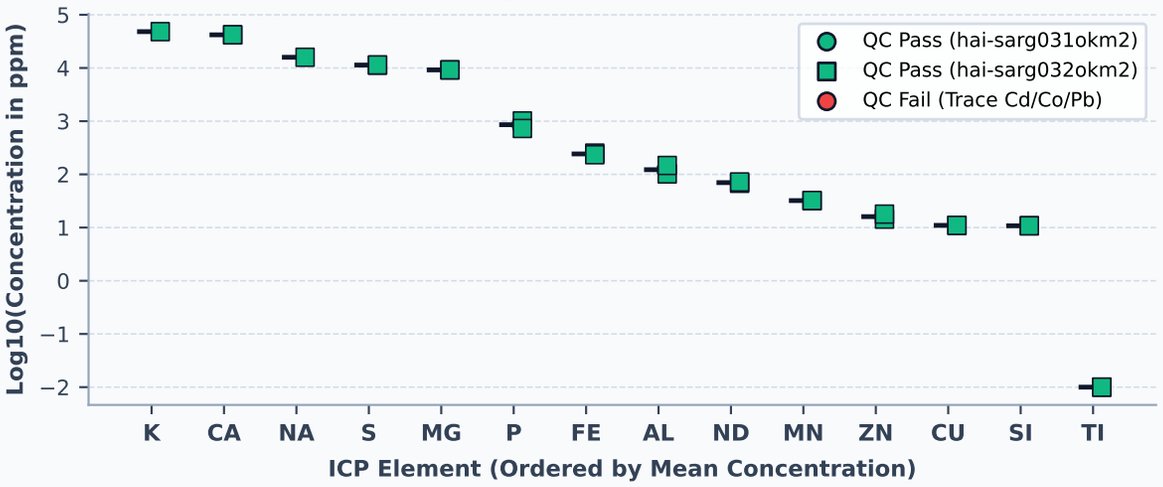
Proximate Analysis — Sample Profile (Scatter + Mean ± SD)



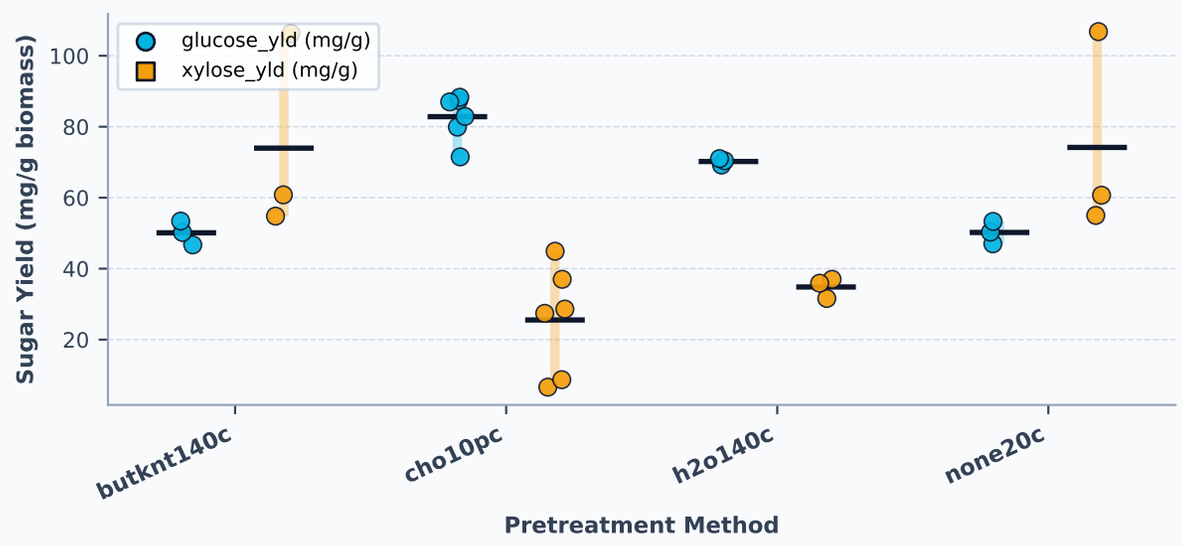
Compositional Analysis — Structural Sugars & Acid-Insol. Lignin



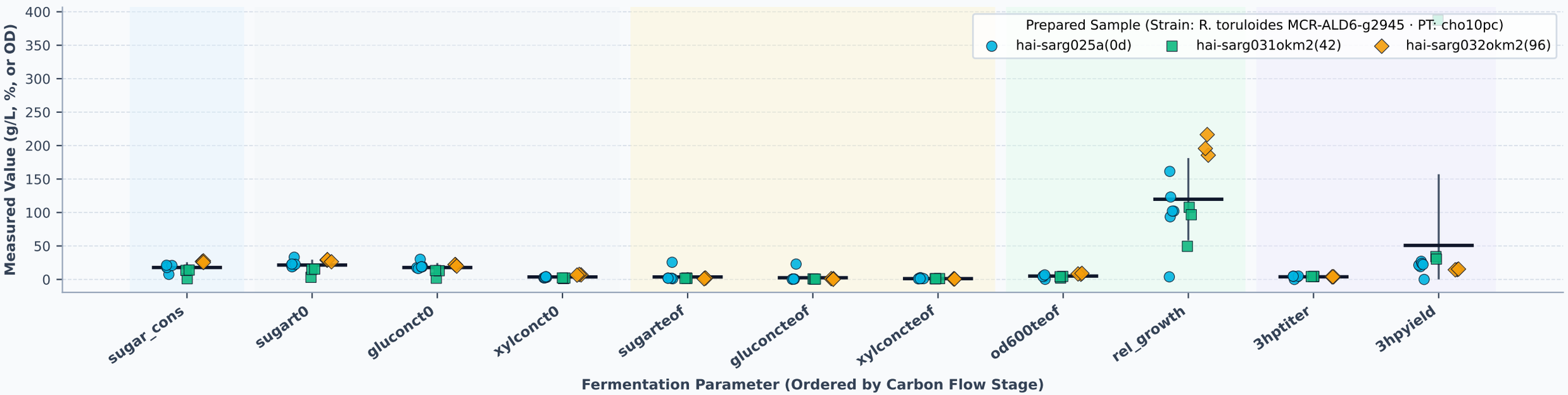
ICP Elemental Profile — Log10 Concentration & QC Status

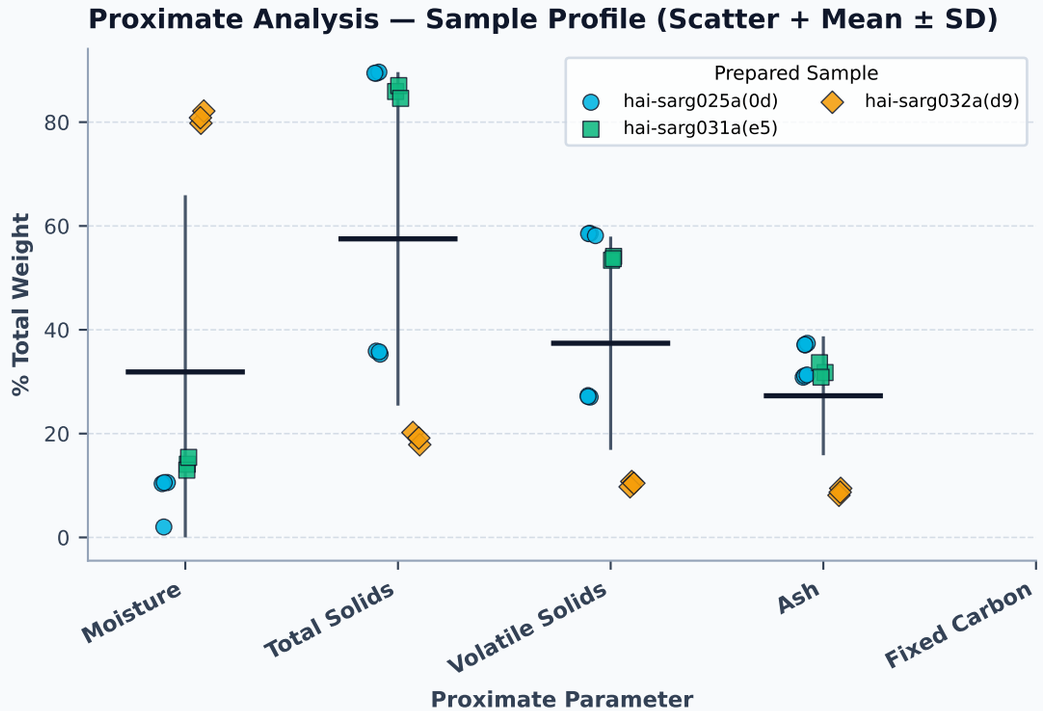


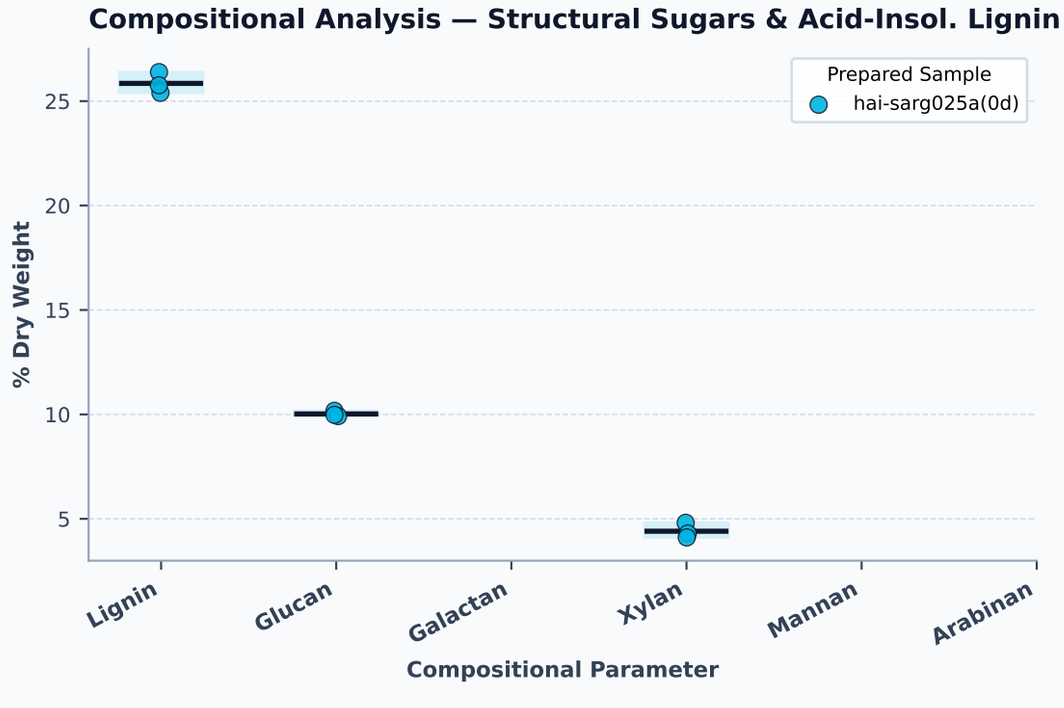
Pretreatment & Hydrolysis — Method Variance (Replicate Strip/Scatter)

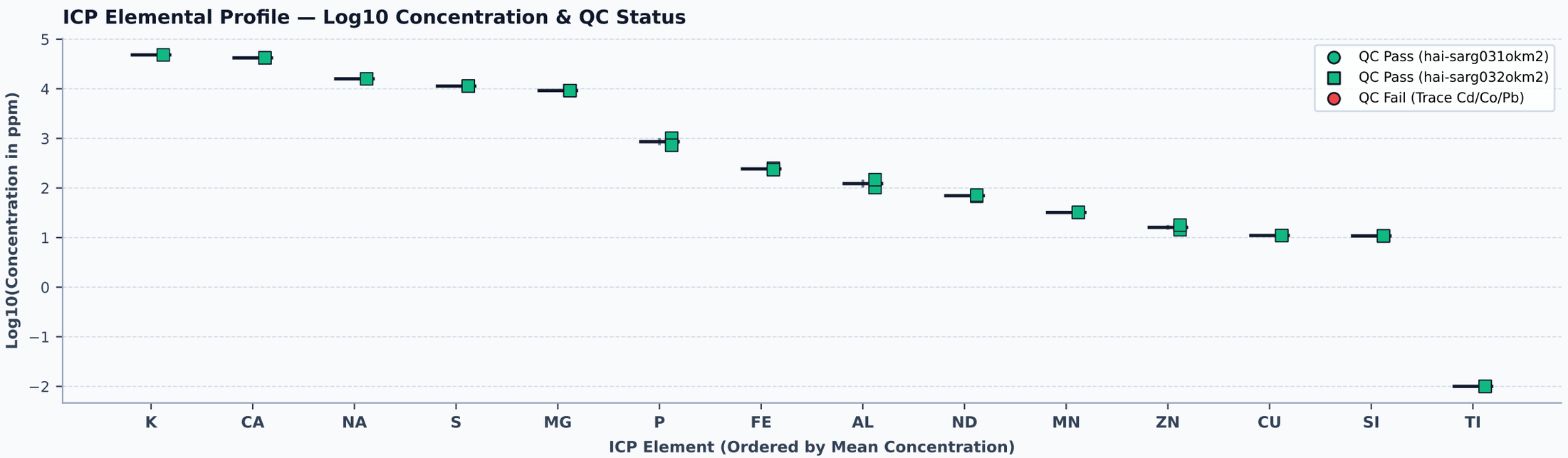


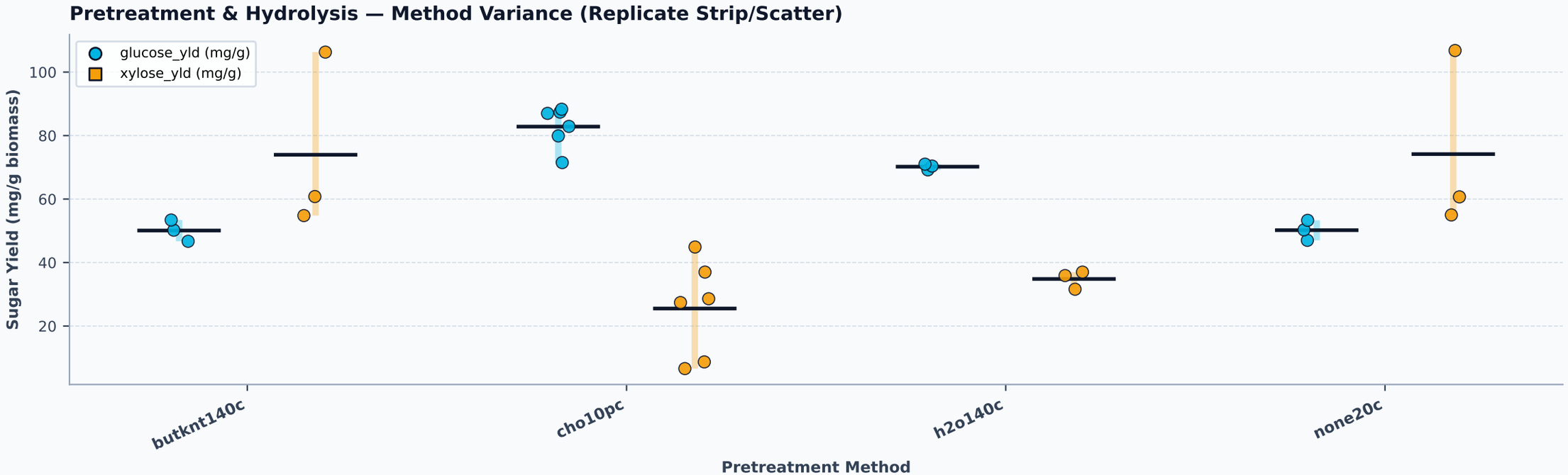
Fermentation (R. toruloides MCR-ALD6-g2945) — Carbon Flow: Sugar Consumption -> T0 -> TEOF -> Growth -> 3-HP & EtOH

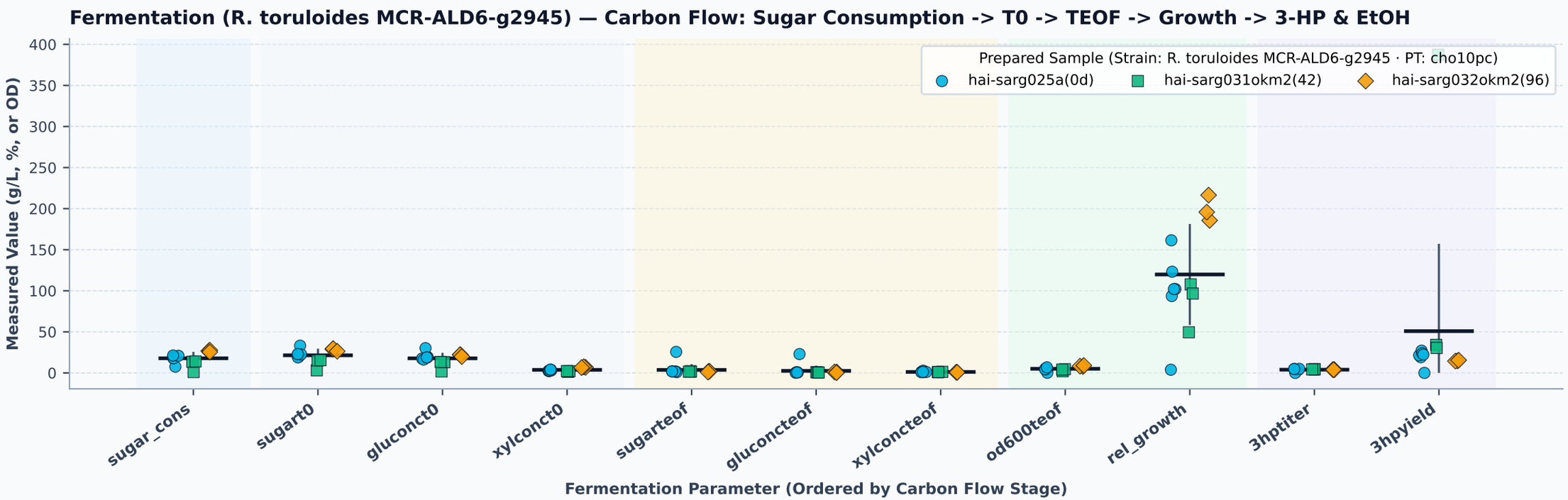



Proximate Analysis — Sample Profile (Scatter + Mean ± SD)
Prepared Sample
hai-sarg025a(0d)
hai-sarg032a(d9)
hai-sarg031a(e5)
% Total Weight
0
20
40
60
80
Moisture
Total Solids
Volatile Solids
Ash
Fixed Carbon
Proximate Parameter


Compositional Analysis — Structural Sugars & Acid-Insol. Lignin
Prepared Sample
hai-sarg025a(0d)
% Dry Weight
5
10
15
20
25
Lignin
Glucan
Galactan
Xylan
Mannan
Arabinan
Compositional Parameter


ICP Elemental Profile — Log10 Concentration & QC Status
QC Pass (hai-sarg031okm2)
QC Pass (hai-sarg032okm2)
QC Fail (Trace Cd/Co/Pb)
Log10(Concentration in ppm)
5
4
3
2
1
0
−1
−2
K
CA
NA
S
MG
P
FE
AL
ND
MN
ZN
CU
SI
TI
ICP Element (Ordered by Mean Concentration)


Pretreatment & Hydrolysis — Method Variance (Replicate Strip/Scatter)
glucose_yld (mg/g)
xylose_yld (mg/g)
Sugar Yield (mg/g biomass)
100
80
60
40
20
butknt140c
cho10pc
h2o140c
none20c
Pretreatment Method


Fermentation (R. toruloides MCR-ALD6-g2945) — Carbon Flow: Sugar Consumption -> T0 -> TEOF -> Growth -> 3-HP & EtOH
Prepared Sample (Strain: R. toruloides MCR-ALD6-g2945 · PT: cho10pc)
hai-sarg025a(0d)
hai-sarg031okm2(42)
hai-sarg032okm2(96)
Measured Value (g/L, %, or OD)
400
350
300
250
200
150
100
50
0
sugar_cons
sugart0
gluconct0
xylconct0
sugarteof
gluconcteof
xylconcteof
od600teof
rel_growth
3hptiter
3hpyield
Fermentation Parameter (Ordered by Carbon Flow Stage)